MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 5795

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, tunable frequency range 3,100 to 3,500 MHz, integral magnet, rated peak power output 1,000 kw

ABSOLUTE RATINGS:

Parameter:	Ef	If(surge)	tpc	rrv	Du	ib	epy
Unit:	V	a	µs	kv/µs	---	a	kv
Maximum:	120	17	1.5	220	0.0022	50	47
Minimum:	---	---	---	---	---	20	---

Parameter:	Pi	pi	VSWR	T(input) bushing	T(body)	Tuner torque	Alt
Unit:	kw	Mw	---	°C	°C	in-lb	ft
Maximum:	5	2.4	3:1	300	110	3	4,000
Minimum:	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Mounting position:	With output choke upwards
Mounting support:	Tube to be mounted on magnet strap feet (see figures 2, 4, and note U)
Marking:	See figures 3 and 4
Magnet:	See note F
Weight:	65.5 lbs (approximate)

TEST CONDITIONS:

Parameter:	Ef	If	tk	Ib	rrv	Du	tpc	VSWR
Unit:	V	A	sec	mAdc	kv/µs	---	µs	---
Tolerance:	±1.0	±0.1	±10	±0.5	±40	±0.0001	±0.08	---
Test condition 1:	115	---	300	---	---	---	---	---
2:	---	2.5	240	90	140	.002	1.33	1.05:1 (max)
3:	---	2.5	240	90	140	.002	1.33	1.50:1 (min)
4:	---	3.3	240	30	140	.000665	1.33	1.50:1 (min)
5:	---	2.5	240	80-90	140	.002	1.33	1.50:1 (min)
6:	---	2.7	240	60	140	---	---	1.05:1 (max)
Notes:	9	18			19,20	14,19	14,19	8

F1 = 3,100 MHz
F2 = 3,300 MHz
F3 = 3,500 MHz
(see note 4)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	TEST CONDITIONS	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1002	Barometric pressure, reduced	2	11	Pressure = 656 mmHg, absolute	---	---	---	---
4309	Phase of sink	---	12	No voltages	---	0.3	0.5	λg
---	Terminal torque, heater	---	13	No voltages	---	---	---	---
4002	Forced convection	3	17	TA = 50°C (max); Pi - Po = 2,250 W Air = 300 cfm (max) through cooling fins for this test only	ΔT Body	---	+60	°C
	Quality conformance inspection, part 1	---	1					
1301	Heater current	1	2		If	3.30	4.00	A
4218	Tunable frequency	6 6	3 3	Upper limit Lower limit	F F	3,505 3,080	3,520 3,100	MHz MHz
4306	Pulse voltage	2	---	F1	epy	43	45	kv
4250	Power output	2	4	F1, F2, F3	Po	2,000	---	W
4315	Stability	3 4	4,5 4,5	F1, F2, F3 F1, F2, F3	MP MP	--- ---	1 1	% %
4308	RF bandwidth	3	4	F1, F2, F3	BW	---	2/tpc	MHz
4308	Minor lobes	3,4	4	F1, F2, F3	Ratio	6	---	dB
4223	Tuner torque or force (operating torque or force)	---	7	No voltages; TA = 25°C	Torque	---	1.5	in./lb
	Quality conformance inspection, part 2							
4223	Tuning rate	2	6	g1 + 1.10 g1 + 5.60 g1 +11.60	F F F	3,410 3,280 3,085	3,455 3,330 3,145	MHz MHz MHz
1031	High-frequency vibration	---	10	No voltages; 10 G; t = 60	---	---	---	---
1042	Shock, specified pulse	---	---	Test condition A	---	---	---	---
4310	Pulling factor	5	4		ΔF	---	10	MHz
4223	Resettability	2	4,15	F1, F3	ΔF	---	5	MHz
	Quality conformance inspection, part 3							
---	Life test	---	16	Group D; hourly cycle	Life	660	---	Cycles
---	Life-test end points:							
4306	Pulse voltage	2	---	F1	epy	41	47	kv
4250	Power output	2	4	F1, F2, F3	Po	1,600	---	W
4315	Stability	3	4,5	F1, F2, F3	MP	---	5	%
4308	Minor lobes	3	4	F1, F2, F3	Ratio	5	---	dB
4218	Tunable frequency	6 6	--- ---	Upper limit Lower limit	F F	3,490 3,070	3,530 3,110	MHz MHz

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NOTES:

1. Un for all tests listed under quality conformance inspection pa el II.
2. The 115 volts for 5 minutes. Then the heater current shall be che ed to 107 volts.
3. The es of the mechanical tuning range. The upper frequency lim es of operating at the high frequency mechanical limit. The hall be checked after 5 minutes of operation at the low fre
4. Th requencies; however, the tube shall be capable of meeting tes n the tunable range.
5. Sta ner position and mismatch phase corresponding to least sta be expressed in terms of the average number of pulses mis ses applied during the period of observation. Missing pul due to both arcing and moding. A pulse is considered to a band ± 1 percent of the normal frequency is less than 70 . Test may be performed during any 5 minute interval of a 3
6. The 75 MHz is designated as g1. The frequency at the other tun mbol column (g1 + dial turns) shall be as specified. Set all igher frequencies. Allow 20 minutes of operation at g1 for llow 5 minutes for tube to reach equilibrium after changing the
7. The oothly over the entire mechanical tuning range when sub nds maximum applied at the drive shaft.
8. The tandard waveguide by means of the choke coupling shown in ough 6.
9. Pri the cathode shall be heated by applying 107 volts to the hea periods, the heater shall be operated at 107 ± 7 volts. Upo , the heater current shall be lowered to the specified va
10. The the magnet strap feet and vibrated perpendicular to the cat strap feet shall be bolted with nonmagnetic bolts to the tab tallic spacer to maintain the required minimum 2-inch cle er magnetic materials. After the vibration test the tube sha shall meet the requirements of quality conformance ins
11. For ut of the tube shall be enclosed in a suitable container whi ressure. This container shall enclose the volume between the t cooling air duct, and the rf shield of the tube.
12. The upled to the magnetron output (phase of sink) is determined by t of the choke coupling (see figure 9) at which a standing wave terms of a fraction of wavelength in the guide (λg). This test tuned to an operational frequency of 3,500 MHz.
13. With to the heater connector (figure 6), apply between 8 and 10 retaining bolt. There shall be no radial or axial force appl be no evidence of a direct short between heater and cath tion to the heater input assembly at the completion of this
14. Appl tition rate and approximately the same shape as required in t itude reduced to obtain the specified Ib. Because of the redu th and duty cycle for this test condition may be slightly less sition 3.

herein
L-E-1.

d note U)

F1 = 3,100 MHz
F2 = 3,300 MHz
F3 = 3,500 MHz
(see note 4)

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NOTES: -Continued

15. The frequency obtained by turning the tuning mechanism to a given setting in one direction shall be reproducible when returning to the same setting from the opposite direction within the specified value.

16. Cycle: 10 minutes at no voltage, 10 minutes at Ef = 107 V, 40 minutes at test condition 2, except σ' = 1.5:1 $^{+15}_{-5}$ percent and with phase varied through 180° at least twice during the pulsing part of the hourly cycle. Frequency shall be changed at the rate of approximately 25 MHz per day, either continuously or in steps.

17. For any ambient temperature to a minimum of 20°C and any anode dissipation to a maximum of 2,250 watts, the tube shall be cooled with air as follows:

Location	Airflow
Body	325 cubic feet per minute (cfm) maximum through fins
Cathode	4.75 ± 0.50 cfm
Tuning head	6.0 ± 0.50 cfm
Output window	1.5 cfm minimum

Body temperature shall be measured by removing one of the six screws holding the output choke and inserting a thermocouple in contact with the bottom of the screw hole. The window shall be cooled through nozzle which directs air stream towards a point 1/4 inch removed from center of the window so that the cooling is developed within a circle of 1/4-inch radius centrally located (see figure 9). Cathode cooling air only is to be provided for test condition 1.

18. There shall be a suitable series impedance during starting time to limit surge current to less than 17 amperes. The cold resistance of the heater is approximately 3.2 ohms.

19. The input circuit should be so designed that the energy per pulse delivered to the tube, if arcing occurs, cannot greatly exceed the normal energy per pulse. If a modulator of the discharge network type is used to satisfy this requirement, the input capacitance as measured at the tube socket with magnetron and pulse transformer removed from the circuit should be 75 ± 15 pF.

20. The rate of rise of voltage (rrv) shall be expressed in kilovolts per microsecond defined as the steepest tangent to the leading edge of the voltage pulse above 80 percent amplitude, where amplitude is defined as a horizontal line drawn tangent to the top of the smooth peak.

21. Changes from previous issue. The extent of changes (deletions, additions, etc.) precludes the annotation of the individual changes from the previous issue of this document.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

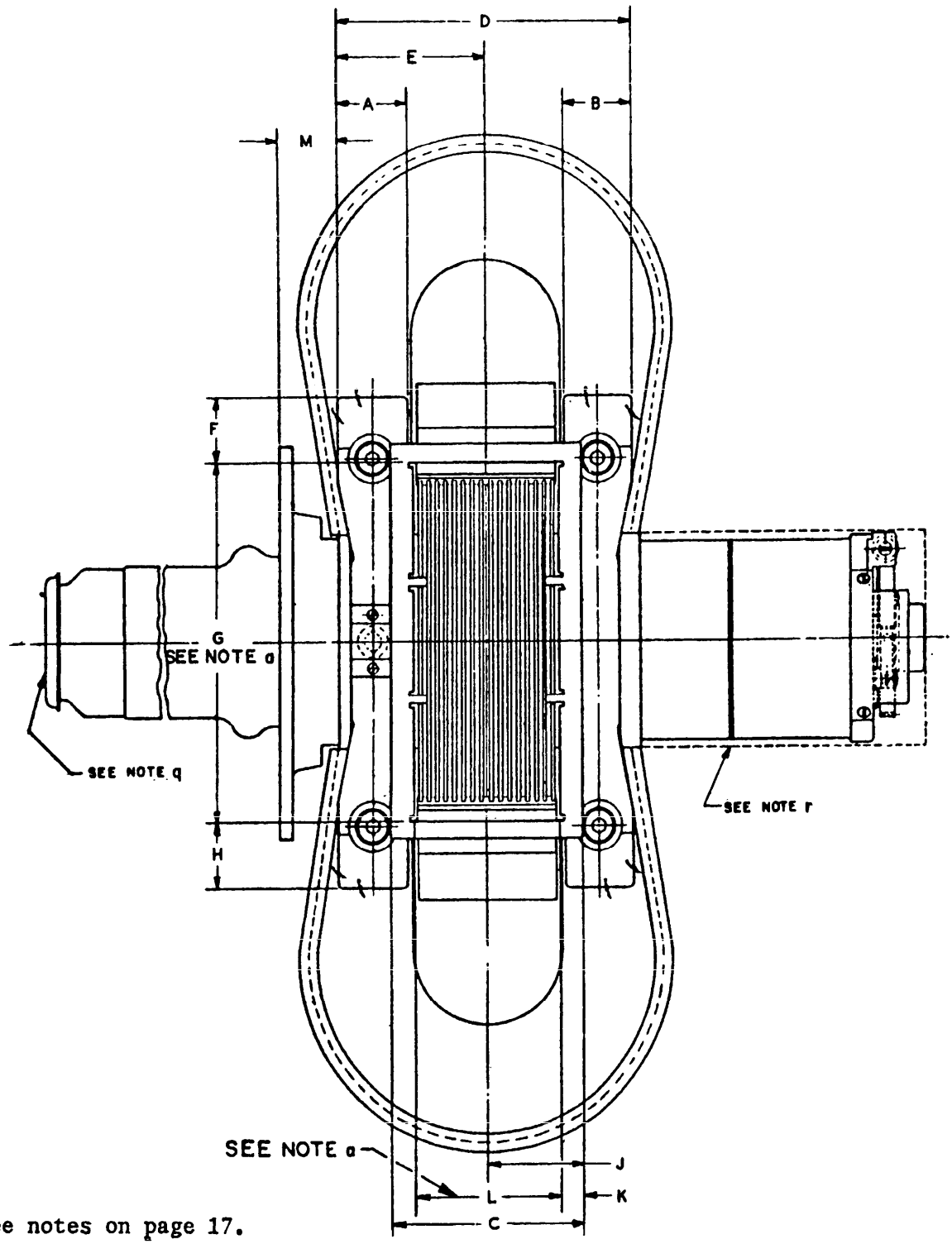
User activities:
Army -
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2845)

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See notes on page 17.

FIGURE 1. Outline drawing of electron tube type 5795.

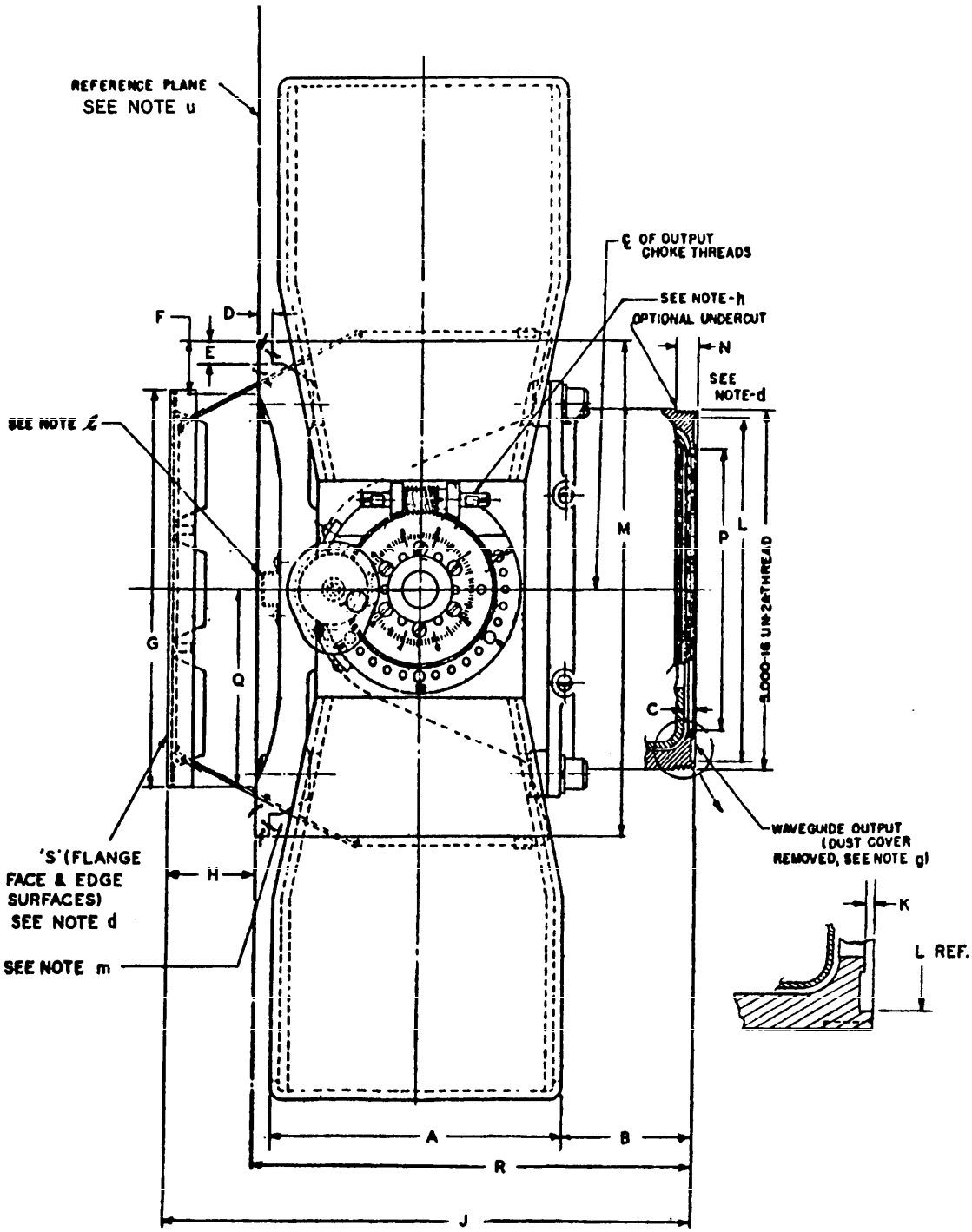
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)		Note
	Minimum	Maximum	
Qualification inspection			
A	.937 (23.80)	1.062 (26.97)	
B	.937 (23.80)	1.062 (26.97)	
C	2.718 (69.04)	2.782 (70.66)	
Quality conformance inspection, part 1			1
M	.781 (19.84)	.905 (22.99)	
Quality conformance inspection, part 2			
D	4.188 (106.38)	4.312 (109.52)	
E	2.063 (32.40)	2.187 (57.15)	
F	.843 (21.41)	.907 (23.04)	
G	4.968 (126.19)	5.032 (127.08)	
H	.843 (21.41)	.907 (23.04)	
J	1.313 (33.35)	1.437 (36.50)	
K	.311 (7.90)	.375 (9.53)	
L	2.030 (51.56)	2.094 (51.19)	

See notes on page 17.

FIGURE 1. Outline drawing of electron tube type 5795 - Continued.

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See notes on page 17.

FIGURE 2. Outline drawing of electron tube type 5795 - Continued.

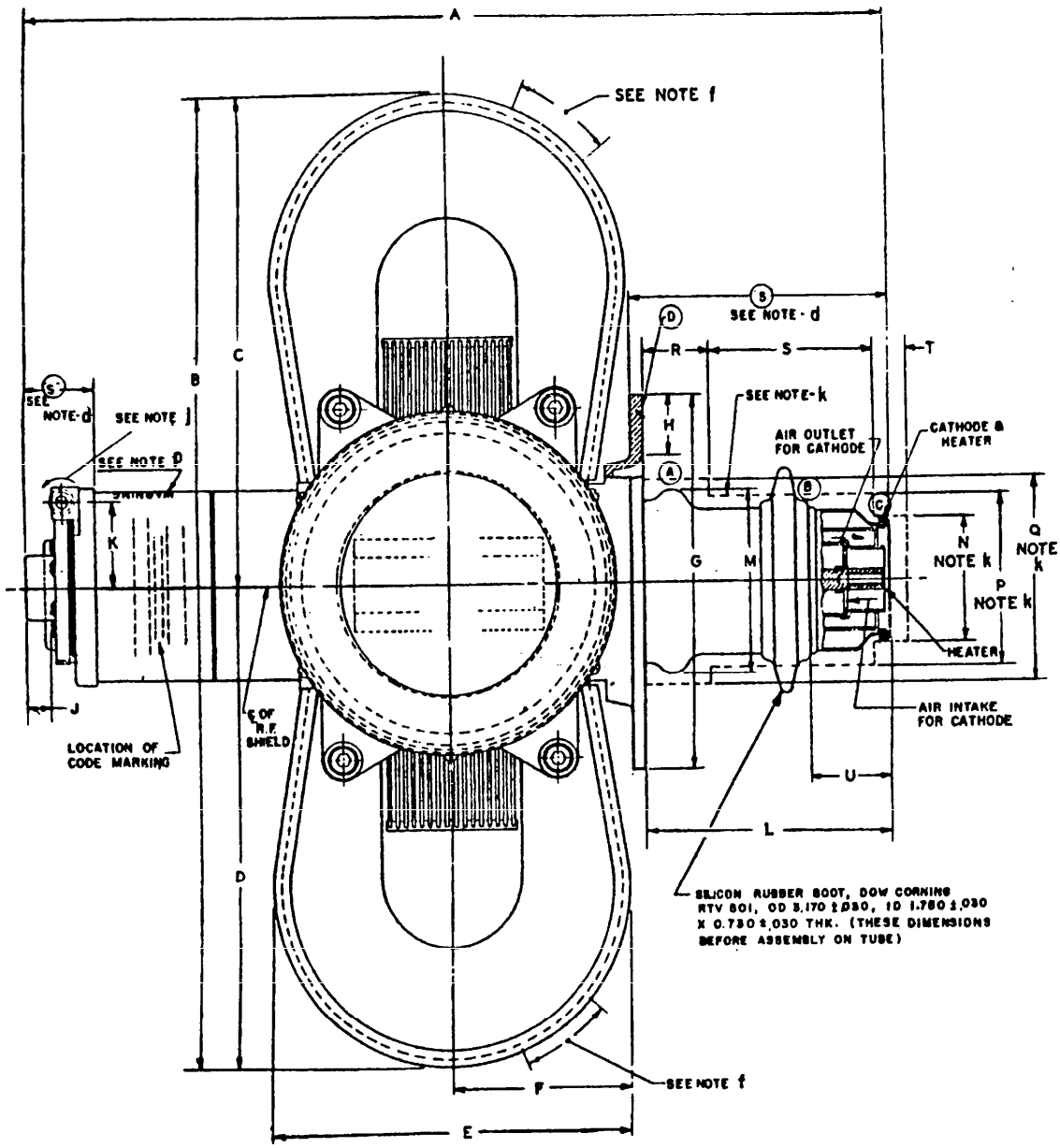
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)		Note
	Minimum	Maximum	
Qualification inspection			
A		4.375 (111.13)	
B	1.750 (44.45)		
C	.130 (3.30)	.182 (4.62)	
Quality conformance inspection, part 1			1
Q	2.688 (68.28)	2.812 (71.42)	
R	6.312 (160.32)	6.352 (161.34)	
Quality conformance inspection, part 2			
D	.187 (4.75)	.249 (6.32)	
E	.218 (5.54)		
F	.468 (11.89)		
G	5.469 (138.91)	5.531 (140.48)	
H	1.188 (30.18)	1.312 (33.32)	
J		7.687 (195.25)	
K	.015 (.38)	.025 (.64)	
L	4.745 (120.52)	4.755 (120.78)	
M	6.735 (171.07)	6.755 (171.58)	
N	.312 (7.92)		
P	3.870 (98.30)	3.880 (98.55)	

See notes on page 17.

FIGURE 2. Outline drawing of electron tube type 5795 - Continued.

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See notes on page 17.

FIGURE 3. Outline drawing of electron tube type 5795 - Continued.

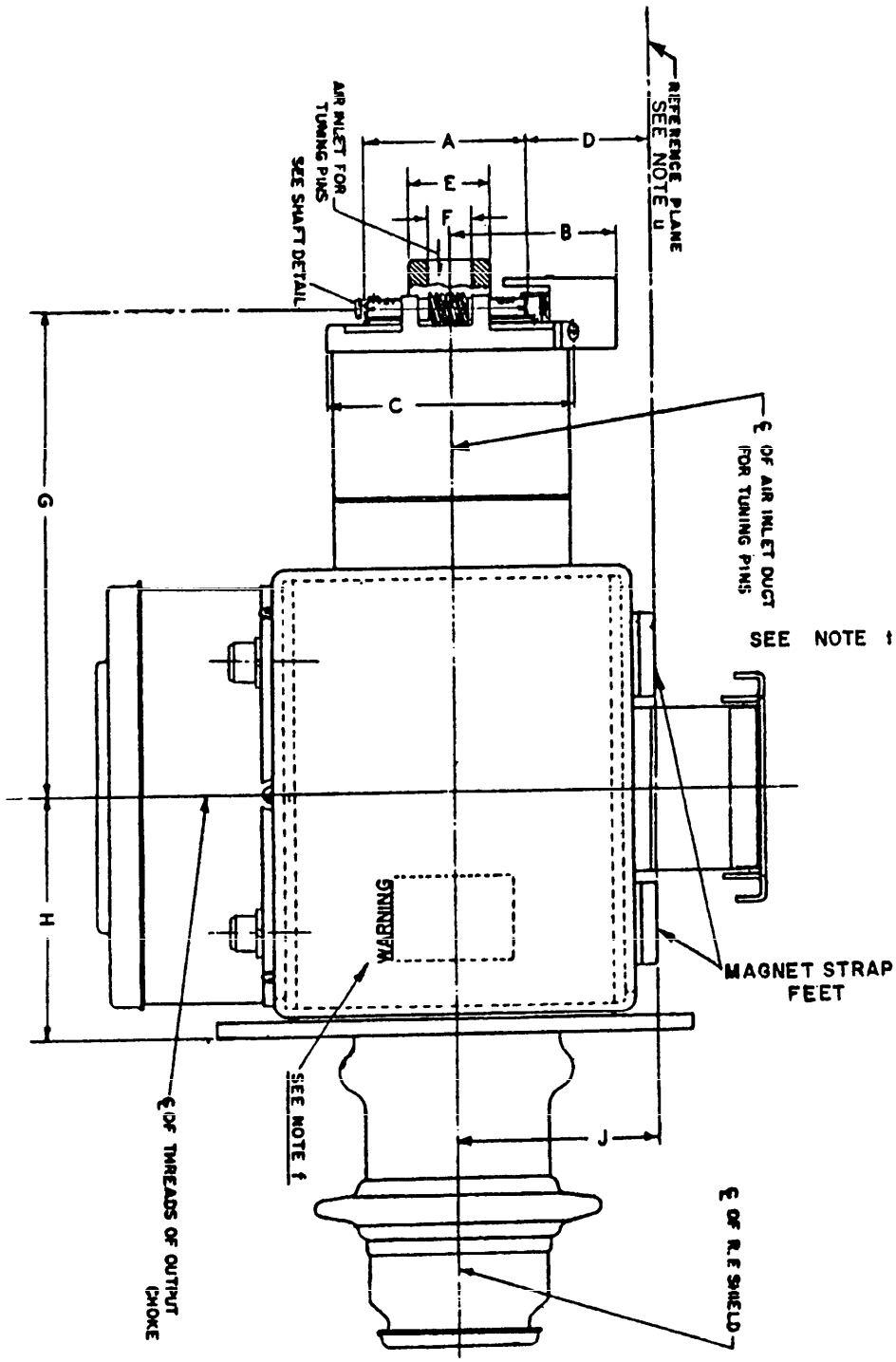
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)		Note
	Minimum	Maximum	
Qualification inspection			
A		13.250 (336.55)	
B		14.500 (368.30)	
C		7.250 (184.15)	
D		7.250 (184.15)	
E		5.625 (142.88)	
F		2.937 (74.60)	
G	5.469 (138.91)	5.531 (140.49)	
H	.750 (19.05)		
J	.312 (7.92)		
Quality conformance inspection, part 1			1
M		2.750 (69.85)	
N		1.800 (45.72)	
P		2.500 (63.50)	
Q		3.000 (76.20)	
Quality conformance inspection, part 2			
K	1.204 (30.58)	1.296 (32.92)	
L	3.688 (93.68)	3.812 (96.82)	
	Reference dimensions		c
R	1.187 (30.15)		
S	2.312 (58.72)		
T	.500 (12.70)		
U	1.250 (31.75)		

See notes on page 17.

FIGURE 3. Outline drawing of electron tube type 5795 - Continued.

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See notes on page 17.

FIGURE 4. Outline drawing of electron tube type 5795 - Continued.

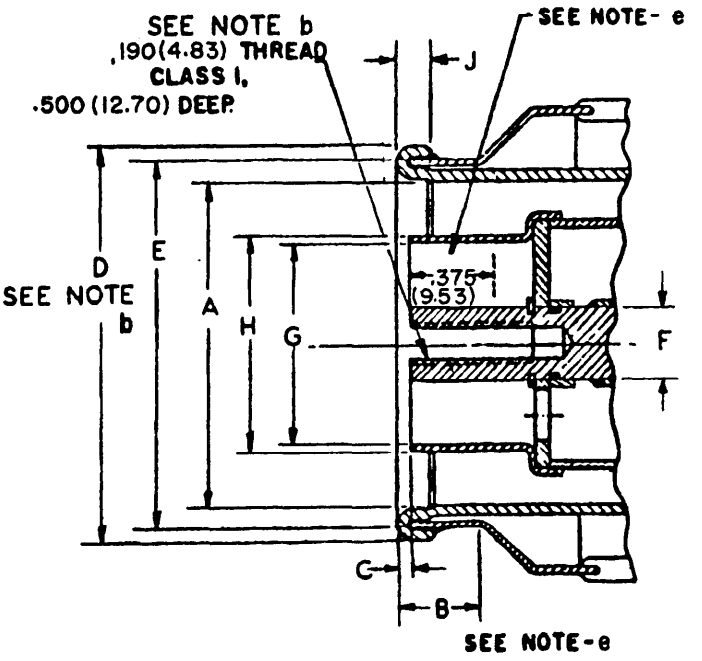
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)	
	Minimum	Maximum
Qualification inspection		
A	1.844 (46.84)	1.906 (48.41)
B		2.062 (52.37)
C		3.000 (76.20)
Quality conformance inspection, part 2		
D	1.422 (36.12)	1.546 (39.27)
E	.921 (23.39)	.951 (24.16)
F	.490 (12.45)	.515 (13.08)
G	5.531 (140.49)	5.655 (143.64)
H	2.922 (74.22)	3.014 (76.56)
J	2.313 (58.75)	2.437 (61.90)

See notes on page 17.

FIGURE 4. Outline drawing of electron tube type 5795 - Continued.

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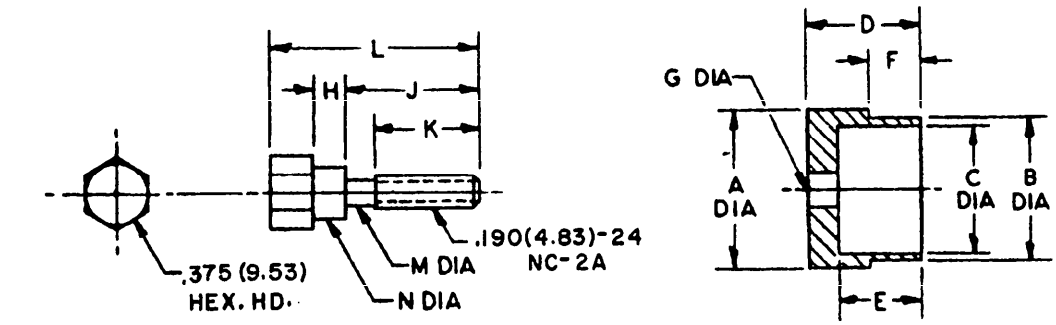
ENLARGED VIEW OF INPUT END
(DUST COVER REMOVED)

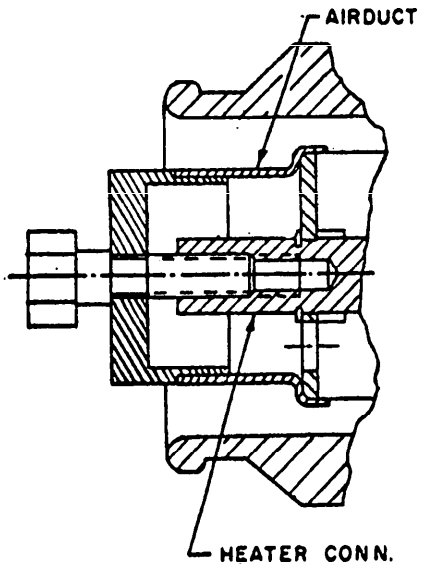
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)		Note
	Minimum	Maximum	
Qualification inspection			
A	1.415 (35.94)	1.455 (36.96)	
B	.266 (6.76)	.390 (9.91)	
C	.031 (.79)	.093 (2.36)	
Quality conformance inspection, part 1			1
G	.873 (22.17)	.879 (22.33)	
Quality conformance inspection, part 2			
D	1.725 (43.82)	1.765 (44.83)	
E	1.620 (41.15)	1.650 (41.91)	
F	.302 (7.67)	.322 (8.18)	
Reference dimensions			c
H	.938 (23.83)		
J	.156 (3.96)		

See notes on page 17.

FIGURE 5. Enlarged view of input end (dust cover removed).

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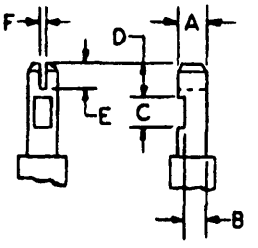


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)	Note
	Nominal dimensions	c
A	.937 (23.81)	
B	.870 (22.12)	
C	.750 (19.05)	
D	.687 (17.46)	
E	.500 (12.70)	
F	.312 (7.94)	
G	.218 (5.54)	
H	.187 (4.76)	
J	.812 (20.64)	
K	.625 (15.88)	
L	1.250 (31.75)	
M	.140 (3.56)	
N	.312 (7.94)	

See notes on page 17.

FIGURE 6. Heater torque gage.

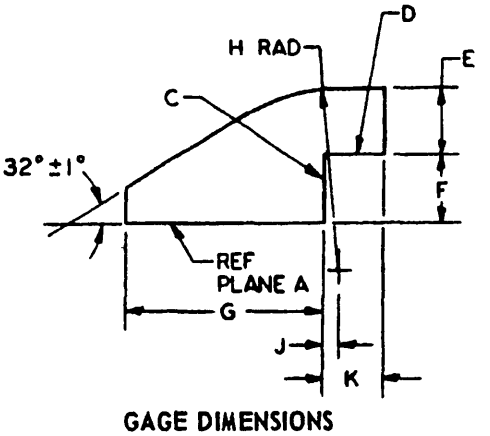
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SHAFT DETAIL
SEE NOTE S

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	.184 (4.67)	.190 (4.83)
B	.146 (3.71)	.166 (4.22)
C	.172 (4.37)	.202 (5.13)
D	.141 (3.58)	.171 (4.34)
E	.110 (2.79)	.140 (3.56)
F	.040 (1.02)	.051 (1.30)

FIGURE 7. Shaft dimensions.



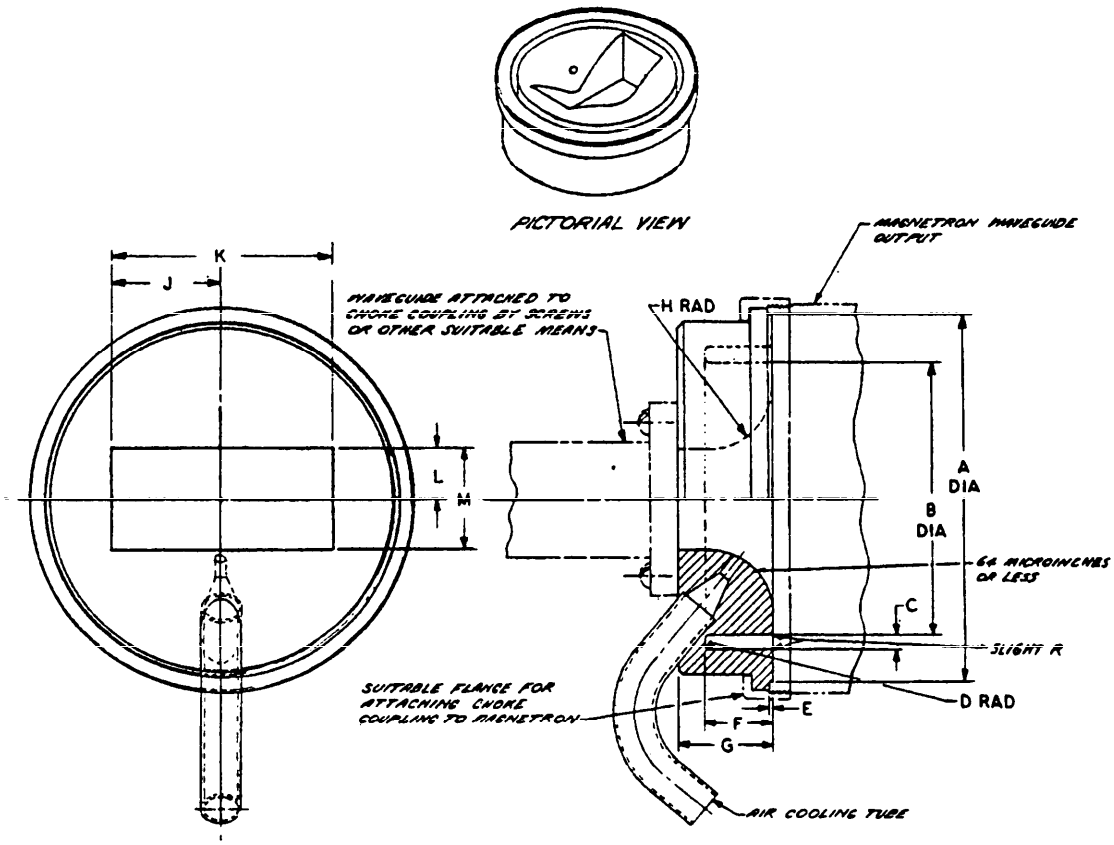
GAGE DIMENSIONS
SEE NOTE m

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)	
	Minimum	Maximum
E	.224 (5.69)	
F	.187 (4.76)	.250 (6.35)
G	.625 (15.88) Nom	
H	.578 (14.68)	
J		.045 (1.14)
K	.172 (4.37)	.203 (5.16)

See notes on page 17.

FIGURE 8. Gage dimensions.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note n)		
	Minimum	Nominal	Maximum
A			4.742 (120.45)
B		3.500 (88.90)	
C		1.880 (47.75)	
D		.055 (1.40)	
E		.031 (.79)	
F		.863 (21.92)	
G	1.199 (30.45)	1.207 (30.66)	1.215 (30.86)
H		.855 (21.72)	
J		1.420 (36.07)	
K		2.840 (72.14)	
L		.670 (17.02)	
M		1.340 (34.04)	

See notes on page 17.

FIGURE 9. Choke coupling.

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NOTES:

a. Dimensions of cooling fin air inlet duct.
b. The .190-24 thread and 1.745 (44.32 mm) diameters to be concentric within .020 (.51 mm).
c. Dimensions without limits are nominal and are for equipment design purposes only and need not be checked.
d. All metal surfaces shall be painted except those marked (S) or those marked (finished surface).
e. No clamping means to bear beyond this depth.
f. Warning:
Maintain minimum clearance of 2 inches (50.80 mm) between this magnet and magnetic materials (magnets, steel tools, plates, etc.).
g. The opening of the waveguide shall be enclosed by the dust cover when tube is not in use.
h. The tuning mechanism shall operate smoothly over the entire mechanical range when subjected to a torque of 1-1/2 inch pounds applied at the drive shaft. The gear and worm threads must be kept free from corrosion, paint and other imperfections.
j. Frequency increases when drive shaft is rotated in the direction shown.
k. The input end of the tube shall fall within the volume limited by three adjacent concentric cylinders (A), (B), and (C) of the dimensions shown, with the axis of the cylinders perpendicular to surface (D).
l. Tubulation cover shall not extend beyond reference plane.
m. A gage 1-inch (25.40 mm) wide having a cross section as shown shall be capable of being inserted in space between tie strap foot and magnet with reference plane 'A' of gage parallel to the reference plane thru the tie strap foot and with surfaces (C) and (D) of gage resting against edge and top of tie strap foot respectively.
n. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
p. Warning:
Handle tube by magnets only. Force applied to other parts may cause damage.
q. The opening of the input end shall be enclosed by the dust cover when not in use.
r. Protective bag over tuning head must be removed before tube is used.
s. Minimum clearance between ends of drive shaft and tuning head bracket shall be .110 inch. (2.79 mm).
t. With the tube resting on a plane surface coincident with the reference plane, .015 (.38 mm) feeler gage shall not enter between tie strap feet and reference plane.
u. The reference plane shall be defined as the plane established by any three of the four magnet tie strap feet.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1522C ELECTRON TUBE, MAGNETRON TYPE 5795

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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